FOUNDATION UNIVERSITY SCHOOL OF SCIENCE & TECHNOLOGY

DEPARTMENT OF SOFTWARE ENGINEERING



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SETION: BSSE-5C

SUBJECT: Software construction and development lab

REQUEST FOR PROPOSAL

PROJECT NAME:

"PROPERTY RENTAL MANAGEMENT SYSTEM"

1. BACKGROUND / INTRODUCTION OF THE SYSTEM

The Property Rental Management System is a centralized software platform designed to simplify the process of finding, listing, renting, and managing residential and commercial properties. In many rental environments, landlords and tenants depend on property dealers, social media groups, messaging applications, classified advertisements, phone calls, and manual records. These methods can make the rental process time-consuming and difficult to track, particularly when a landlord manages multiple properties.

The proposed system will provide a single digital platform through which landlords can create and manage property listings, tenants can search and apply for available properties, and administrators can verify listings and manage users. The system will also support rental applications, rental agreements, rent records, maintenance requests, notifications, reviews, and reporting.

The project is intended to demonstrate practical software development and construction concepts, including requirements engineering, object-oriented design, layered architecture, database management, authentication and authorization, validation, testing, error handling, and maintainable software design.

2. PROBLEM STATEMENT

The conventional property rental process is often fragmented. Tenants may need to search through multiple sources to find a suitable property, while landlords may manage property information, tenant applications, rent payments, and maintenance requests using separate manual or digital methods.

- Property listings may be incomplete, duplicated, outdated, or difficult to verify.
- Tenants may spend considerable time searching for properties that match their budget and requirements.
- There may be no standardized method for submitting and tracking rental applications.
- Rental agreements and tenant information may be stored manually and become difficult to retrieve.
- Landlords may have difficulty tracking monthly rent, late payments, and outstanding amounts.
- Maintenance complaints may be communicated through informal channels and may not be properly tracked.
- There may be limited transparency regarding property availability and rental status.
- Administrators may lack a centralized view of users, listings, complaints, and rental activity.

Therefore, a centralized Property Rental Management System is proposed to organize the complete rental workflow and reduce manual effort while improving communication, record keeping, and transparency.

3. BUSINESS OBJECTIVES AND PRELIMINARY REQUIREMENTS (SRS)

3.1 Business Objectives

1. Provide a centralized platform for landlords to advertise and manage rental properties.
2. Allow tenants to search and filter properties according to their requirements.
3. Reduce the time and effort involved in finding and renting a property.
4. Digitize rental applications and rental agreement management.
5. Maintain accurate records of rent payments, due dates, and outstanding amounts.
6. Provide a structured process for reporting and resolving maintenance issues.
7. Improve communication through system notifications and status updates.
8. Provide administrators with verification, monitoring, reporting, and user-management capabilities.
9. Create a scalable foundation that can later support online payments, maps, and other integrations.

3.2 Main Functional Requirements

The following are the main preliminary functional requirements for the initial project scope. Additional functional requirements will be identified, refined, and added during requirements elicitation, analysis, and validation as development progresses.

ID	REQUIREMENT	DESCRIPTION
FR-01	User Registration	The system shall allow landlords, tenants, and authorized staff to create user accounts using required personal and contact information.
FR-02	Authentication	The system shall allow registered users to securely log in and log out.
FR-03	Role-Based Access	The system shall provide different permissions for administrators, landlords, tenants, and authorized maintenance staff.
FR-04	Property Listing	The system shall allow landlords to add property listings containing title, description, location, property type, rent, deposit, bedrooms, bathrooms, area, amenities, availability, and images.
FR-05	Property Management	The system shall allow landlords to edit, deactivate, reactivate, or remove their own property listings according to system rules.
FR-06	Property Search and Filtering	The system shall allow tenants and visitors to search and filter properties using relevant criteria such as location, rent range, property type, bedrooms, bathrooms, amenities, furnishing status, and availability.
FR-07	Property Details	The system shall display detailed information, images, amenities, rental price, availability, and relevant

		landlord information for an approved property.
FR-08	Property Verification	The system shall allow authorized administrators to review and approve or reject property listings before publication when verification is required.
FR-09	Rental Application	The system shall allow authenticated tenants to submit applications for available properties.
FR-10	Application Tracking and Decision	The system shall allow tenants to view application status and authorized landlords to review, approve, or reject applications.
FR-11	Rental Agreement	The system shall allow an approved rental application to be converted into a rental agreement containing tenant, landlord, property, rent, deposit, dates, and payment terms.
FR-12	Rent Management	The system shall record rent obligations, due dates, payments, late payments, outstanding amounts, and relevant payment history.
FR-13	Maintenance Requests	The system shall allow tenants to submit maintenance requests and authorized users to assign and track them through defined statuses.
FR-14	Notifications	The system shall generate notifications for important events such as application decisions, rent due dates, maintenance updates, and agreement changes.
FR-15	Reports and Activity Records	The system shall provide authorized users with relevant reports and maintain important activity records for accountability.

3.3 Main Non-Functional Requirements

The following are the main preliminary non-functional requirements. Further quality attributes and measurable constraints will be refined during the later requirements and design phases.

ID	QUALITY ATTRIBUTE	REQUIREMENT
NFR-01	Security	The system shall protect user accounts and sensitive information through authentication, authorization, password protection, and input validation.
NFR-02	Performance	Common operations such as login, property search, filtering, and viewing property details shall respond within an acceptable time under normal project usage.
NFR-03	Usability	The interface shall be simple, consistent, and understandable for users with basic computer or smartphone skills.
NFR-04	Reliability	The system shall preserve important property, application, agreement, and payment records accurately and consistently.
NFR-05	Maintainability	The software shall use modular components, clear naming, documentation, and separation of

		responsibilities so future changes can be made easily.
NFR-06	Data Integrity and Privacy	The system shall validate data and apply appropriate database constraints while restricting users to the personal and rental information for which they have permission.
NFR-07	Compatibility	The web interface shall function correctly on commonly used modern browsers and different screen sizes.
NFR-08	Backup and Recovery	Important database information shall be backed up regularly and a recovery procedure shall be defined.
NFR-09	Testability	Core business logic shall be structured so that unit and integration testing can be performed.
NFR-10	Error Handling	The system shall display understandable error messages and handle invalid input without crashing.

4. CURRENT SITUATION AND OPPORTUNITY STATEMENT

4.1 Current Situation

Property rental activities are commonly distributed across property dealers, classified advertisements, social media groups, messaging applications, phone calls, spreadsheets, and paper documents. These methods are useful for individual transactions but can become inefficient when many properties and tenants are involved.

- Tenants may have to contact several sources before finding a suitable property.
- Property information can become outdated when availability changes.
- Landlords may maintain separate records for properties, tenants, rent, and maintenance.
- Communication history may be scattered across phone calls and messaging platforms.
- There may be no standard process for verifying listings and rental applications.

4.2 Opportunity Statement

A digital rental platform creates an opportunity to combine property discovery and property management in one system. It can provide searchable listings, standardized rental workflows, organized records, notifications, and dashboards. In later versions, the system could be extended with maps, online payment gateways, digital signatures, automated reminders, and property recommendation features.

5. OPTIONS AND RECOMMENDATIONS

Option 1: Manual / Social Media-Based Process

Landlords continue using social media, messaging groups, classified advertisements, property dealers, and phone calls.

Advantages: familiar, low technical setup, and easy for individual users.

Disadvantages: information is scattered, searching is difficult, records are inconsistent, and there is limited verification and rental management.

Option 2: Basic Property Listing Website

A website would allow landlords to publish properties and tenants to search and contact them.

Advantages: centralized listings, easier search, and simpler implementation.

Disadvantages: limited support for agreements, rent records, maintenance, notifications, and long-term management.

Option 3: Complete Property Rental Management System

A complete platform would cover property discovery, user roles, property verification, applications, rental agreements, rent records, maintenance, notifications, reviews, and dashboards.

Advantages: provides an end-to-end rental workflow and strong opportunities to demonstrate software design and construction principles.

Disadvantages: larger scope and requires careful planning, testing, and phased development.

Recommendation

Option 3 is proposed for the academic project because it provides a meaningful real-world problem and sufficient scope to demonstrate requirements analysis, object-oriented programming, database design, architecture, design patterns, security, testing, and project management. The system can be developed incrementally, with core features completed first and advanced management features completed in later phases.

6. SCHEDULE ESTIMATE

WEEK	ACTIVITIES	MILESTONE
1	Topic finalization, background research, problem analysis	Project scope
2	Requirements gathering and SRS preparation	SRS draft
3	Use cases, UML, architecture and workflow design	Design
4	Database design and UI prototype	ERD + prototype
5	Authentication, registration and role management	User module
6	Property listing and management	Property module
7	Search, filtering and rental applications	Core workflow
8	Integration, testing and presentation	MIDTERM
9	Rental agreements and tenant management	Agreement module
10	Rent records, due dates and receipts	Payment module

11	Maintenance requests and assignment	Maintenance module
12	Notifications, reviews and verification	Advanced features
13	Admin dashboard, reports and security improvements	Admin module
14	Unit/integration testing and bug fixing	Testing
15	Final integration, documentation and deployment preparation	Release candidate
16	Final testing, presentation and demonstration	FINAL

7. POTENTIAL RISKS AND MITIGATION PLAN

RISK	IMPACT	MITIGATION
Scope becomes too large	High	Prioritize core requirements and implement advanced features in phases.
Fake or inaccurate property listings	High	Use admin verification, reporting, and listing-status controls.
Unauthorized access	High	Use authentication, role-based authorization, validation, and protected routes.
Payment record errors	High	Validate transactions, use database constraints, and keep backups.
Data loss	High	Perform regular backups and document recovery procedures.
Incorrect property availability	Medium	Update availability based on approved applications and active agreements.
Maintenance requests are missed	Medium	Use request IDs, assignment, status tracking, and notifications.
Development delays	Medium	Use weekly milestones, task allocation, version control, and early testing.
Security vulnerabilities	High	Validate inputs, protect credentials, test authorization, and avoid exposing sensitive data.
Poor user experience	Medium	Use a simple responsive UI and conduct usability testing with sample users.

8. PRELIMINARY SCOPE AND FUTURE REQUIREMENT DEVELOPMENT

At this project-initiation stage, the Business Case Charter contains the main business objectives and the main preliminary functional and non-functional requirements. The complete SRS will be developed progressively as the project moves through the requirements engineering process. Requirements will be elicited, analyzed, validated, documented, and managed as the system